

# An Analysis of DBSCAN

DBSCAN

## DBSCAN -- Part 1

Disadvantages DBSCAN is not entirely deterministic: border points that are reachable from more than one cluster can be part of either cluster, depending on the order the data are processed. For most data sets and domains, this situation does not arise often and has little impact on the clustering result: both on core points and noise points, DBSCAN is deterministic. DBSCAN\* is a variation that treats border points as noise, and this way achieves a fully deterministic result as well as a more consistent statistical interpretation of density-connected components. The quality of DBSCAN depends on the distance measure used in the function  $\text{regionQuery}(P, \epsilon)$ . The most common distance metric used is Euclidean distance. Especially for high-dimensional data, this metric can be rendered almost useless due to the so-called "Curse of dimensionality", making it difficult to find an appropriate value for  $\epsilon$ . This effect, however, is also present in any other algorithm based on Euclidean distance. DBSCAN cannot cluster data sets well with large differences in densities, since the  $\text{minPts}-\epsilon$  combination cannot then be chosen appropriately for all clusters. If the data and scale are not well understood, choosing a meaningful distance threshold  $\epsilon$  can be difficult. See the section below on extensions for algorithmic modifications to handle these issues.

## DBSCAN -- Part 2

Extensions Generalized DBSCAN (GDBSCAN) is a generalization by the same authors to arbitrary "neighborhood" and "dense" predicates. The  $\epsilon$  and  $\text{minPts}$  parameters are removed from the original algorithm and moved to the predicates. For example, on polygon data, the "neighborhood" could be any intersecting polygon, whereas the density predicate uses the polygon areas instead of just the object count. Various extensions to the DBSCAN algorithm have been proposed, including methods for parallelization, parameter estimation, and support for uncertain data. The basic idea has been extended to hierarchical clustering by the OPTICS algorithm. DBSCAN is also used as part of subspace clustering algorithms like PreDeCon and SUBCLU. HDBSCAN\* is a hierarchical version of DBSCAN which is also faster than OPTICS, from which a flat partition consisting of the most prominent clusters can be extracted from the hierarchy.

## DBSCAN -- Part 3

MinPts: As a rule of thumb, a minimum  $\text{minPts}$  can be derived from the number of dimensions  $D$  in the data set, as  $\text{minPts} \geq D + 1$ . The low value of  $\text{minPts} = 1$  does not make sense, as then every point is a core point by definition. With  $\text{minPts} \leq 2$ , the result will be the same as of hierarchical clustering with the

single link metric, with the dendrogram cut at height  $\epsilon$ . Therefore,  $\text{minPts}$  must be chosen at least 3. However, larger values are usually better for data sets with noise and will yield more significant clusters. As a rule of thumb,  $\text{minPts} = 2 \cdot \text{dim}$  can be used, but it may be necessary to choose larger values for very large data, for noisy data or for data that contains many duplicates.  $\epsilon$ : The value for  $\epsilon$  can then be chosen by using a  $k$ -distance graph, plotting the distance to the  $k = \text{minPts} - 1$  nearest neighbor ordered from the largest to the smallest value. Good values of  $\epsilon$  are where this plot shows an "elbow": if  $\epsilon$  is chosen much too small, a large part of the data will not be clustered; whereas for a too high value of  $\epsilon$ , clusters will merge and the majority of objects will be in the same cluster. In general, small values of  $\epsilon$  are preferable, and as a rule of thumb only a small fraction of points should be within this distance of each other. Alternatively, an OPTICS plot can be used to choose  $\epsilon$ , but then the OPTICS algorithm itself can be used to cluster the data. Distance function: The choice of distance function is tightly coupled to the choice of  $\epsilon$ , and has a major impact on the results. In general, it will be necessary to first identify a reasonable measure of similarity for the data set, before the parameter  $\epsilon$  can be chosen. There is no estimation for this parameter, but the distance functions needs to be chosen appropriately for the data set. For example, on geographic data, the great-circle distance is often a good choice. OPTICS can be seen as a generalization of DBSCAN that replaces the  $\epsilon$  parameter with a maximum value that mostly affects performance.  $\text{MinPts}$  then essentially becomes the minimum cluster size to find. While the algorithm is much easier to parameterize than DBSCAN, the results are a bit more difficult to use, as it will usually produce a hierarchical clustering instead of the simple data partitioning that DBSCAN produces. Recently, one of the original authors of DBSCAN has revisited DBSCAN and OPTICS, and published a refined version of hierarchical DBSCAN (HDBSCAN\*), which no longer has the notion of border points. Instead, only the core points form the cluster.

## DBSCAN -- Part 4

A point  $p$  is a core point if at least  $\text{minPts}$  points are within distance  $\epsilon$  of it (including  $p$ ). A point  $q$  is directly reachable from  $p$  if point  $q$  is within distance  $\epsilon$  from core point  $p$ . Points are only said to be directly reachable from core points. A point  $q$  is reachable from  $p$  if there is a path  $p_1, \dots, p_n$  with  $p_1 = p$  and  $p_n = q$ , where each  $p_{i+1}$  is directly reachable from  $p_i$ . Note that this implies that the initial point and all points on the path must be core points, with the possible exception of  $q$ . All points not reachable from any other point are outliers or noise points. Now if  $p$  is a core point, then it forms a cluster together with all points (core or non-core) that are reachable from it. Each cluster

contains at least one core point; non-core points can be part of a cluster, but they form its "edge", since they cannot be used to reach more points.

### DBSCAN -- Part 5

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**Parameter estimation** Every data mining task has the problem of parameters. Every parameter influences the algorithm in specific ways. For DBSCAN, the parameters  $\epsilon$  and minPts are needed. The parameters must be specified by the user. Ideally, the value of  $\epsilon$  is given by the problem to solve (e.g. a physical distance), and minPts is then the desired minimum cluster size.